

Lines, Triangles & Polygons



Parallel lines and transversals

- 1 Two parallel lines are cut by a transversal, forming an angle of 105° on one line. Find the corresponding angle, the alternate interior angle, and the co-interior (same-side interior) angle on the other line.
 - 2 Two angles formed by a transversal crossing parallel lines are alternate exterior angles, so they are equal. If one measures $3x + 10$ and the other measures $5x - 30$, find x and the measure of the angle.
 - 3 Two parallel lines are cut by a transversal. One interior angle is $4x + 15$, and its corresponding angle on the other line is $6x - 25$. Find x and the angle measure.
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Triangle angle sum and the exterior angle theorem

- 4 The angles of a triangle are in the ratio $3 : 5 : 7$. Find each angle.
 - 5 In triangle ABC , $\angle A = 52^\circ$ and the exterior angle at C is 130° . Find $\angle B$.
 - 6 An exterior angle of a triangle measures 115° , and one remote interior angle is 50° . Find the other remote interior angle and the third (adjacent) interior angle.
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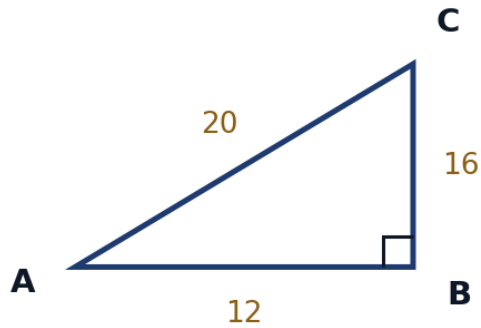
Triangle congruence criteria

- 7 State which congruence criterion (SSS, SAS, ASA, AAS, or RHS) proves triangles PQR and XYZ congruent, given $PQ = XY$, $\angle Q = \angle Y$, and $QR = YZ$.
 - 8 Triangle DEF has $DE = 9$, $EF = 12$, $\angle E = 90^\circ$, and triangle GHI has $GH = 9$, $HI = 12$, $\angle H = 90^\circ$. Are the triangles congruent? By which criterion?
 - 9 Two triangles have all three pairs of corresponding sides equal in length. Which congruence criterion applies, and are the triangles necessarily congruent?
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Similar triangles and proportional reasoning

- 10 Triangle $ABC \sim$ triangle DEF , with $AB = 6$, $BC = 8$, and $DE = 9$. Find EF .

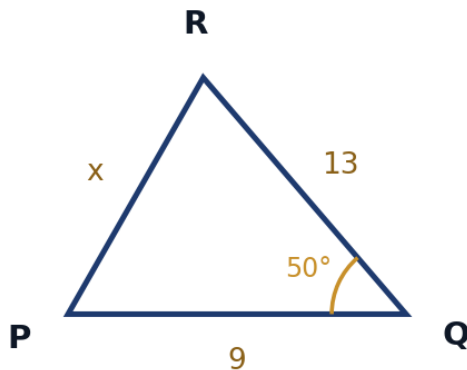
- 11 Triangle ABC shown below has $AB = 12$, $BC = 16$, and $CA = 20$. It is similar to triangle DEF , where $DE = 15$ corresponds to AB . Find EF and FD .



- 12 Triangle JKL has $JK = 15$ and $KL = 21$, and is similar to triangle MNO with scale factor $\frac{2}{3}$ (triangle MNO is smaller), where MN corresponds to JK . Given $MN = 10$, verify the scale factor is consistent, then find NO .

The Law of Cosines

- 13 In triangle PQR shown below, $PQ = 9$, $QR = 13$, and $\angle Q = 50^\circ$. Find $PR = x$, to the nearest hundredth.



- 14 A triangle has sides $a = 7$, $b = 8$, $c = 10$. Use the Law of Cosines to find angle C (opposite side c), to the nearest tenth of a degree.
- 15 A triangle has sides $a = 6$, $b = 6$, and included angle $C = 80^\circ$. Find side c , to the nearest hundredth, and state what type of triangle this is based on the two given equal sides.

Heron's formula

- 16 Find the area of a triangle with sides 13, 14, 15 using Heron's formula.
- 17 A triangular plot of land has sides 9, 10, 17. Find its area using Heron's formula.

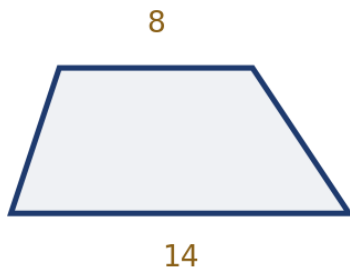
- 18 A triangular sail has sides 5 m, 12 m, and 13 m. Find its area using Heron's formula, and verify using the right-triangle area formula (since 5-12-13 is a Pythagorean triple).
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Interior and exterior angles of polygons

- 19 Find the sum of the interior angles of a regular decagon (10 sides), and the measure of each interior angle.
- 20 Each exterior angle of a regular polygon measures 24° . How many sides does the polygon have?
- 21 A regular polygon has an interior angle of 150° . How many sides does it have?
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Area of polygons

- 22 The trapezoid below has parallel sides of length 14 and 8, and height 6. Find its area.



- 23 Find the area of a regular hexagon with side length 6.
- 24 A composite figure is an 8-by-10 rectangle with a 3-by-4 rectangular notch removed from one corner. Find the remaining area.
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A well-designed word problem: a surveyor's triangular lots

- 25 This question has three parts. A surveyor measures a triangular lot with sides 30 m, 40 m, and 50 m. (a) Verify using the Pythagorean theorem that the lot is a right triangle. (b) Find the area of the lot using the right-triangle area formula, then confirm your answer using Heron's formula. (c) A second, similar triangular lot has its shortest side measuring 45 m. Using the similarity ratio, find the area of the second lot.